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Responsible Design, Development, & Deployment of Technologies: Cultivating Ethical & Trustworthy Artificial Intelligence in US Food Systems

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Executive Summary
Artificial Intelligence (AI) is driving the creation and rapid advancement of transformative tools across the agricultural value chain, with potential to support climate resilience, sustainability, nutrition security, and equitable access to food. However, there are significant concerns around the ethical, social, and legal dimensions of AI that must be addressed prior to the widespread adoption of these technologies in food systems. To begin addressing these concerns, a series of convenings for food systems stakeholders were held in early 2025, sponsored through the National Science Foundation's Responsible Design, Development, and Deployment of Technologies (ReDDDoT) program. The aim of these convenings was to build consensus around the challenges, priorities, and best practices for successfully and equitably adopting AI in food systems. Key needs identified by the stakeholders included:

- Expanding equitable access to AI technologies and infrastructure in rural and underserved communities.
- Strengthening trust, transparency, and data stewardship to ensure responsible data collection and innovation.
- Developing proactive and harmonized regulatory frameworks at state and federal levels.
- Building workforce readiness and education pathways to support AI literacy and adoption.
- Encouraging collaboration and user-centered design to align AI tools with real-world food system needs.

Stakeholders emphasized the importance of collaborative approaches that prioritize inclusivity, transparency, and user-centered design. Education and workforce training are needed to develop AI literacy, while policy frameworks must provide safeguards without stifling innovation. Ultimately, responsible AI adoption will depend on coordinated efforts across policymakers, developers, researchers, and end-users to ensure the technology delivers sustainable, equitable benefits for U.S. communities and food systems. Here, we document the outcomes of the convenings and include a finalized description of stakeholder recommendations for policy, regulation, education and workforce development, and stakeholder engagement through the technology life cycle.

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Introduction
Today's food systems face mounting pressure to feed a growing global population while grappling with challenges in sustainability, climate resilience, nutrition, access, and the transition to a circular bioeconomy. Artificial intelligence (AI) offers transformative potential across the US food and agriculture sector, spanning the entire value chain from "seed to stomach". By enhancing productivity, efficiency, and environmental stewardship, AI can support sustainable practices, strengthen resilience to climate change, and expand equitable access to nutritious food. To ensure AI is effectively integrated into the US food systems, it is essential to address critical issues around workforce training, education, policy and user acceptance.

Artificial Intelligence: A transformative tool for improving food systems
AI has incredible potential to transform the food system, contribute to climate change mitigation, and aid in the elimination of nutrition insecurity and diet-related disease (1). Artificial intelligence is generally understood to be algorithms that appear to mimic human intelligence (2), with generative capabilities and capacity to improve automation. Ongoing applications of AI in food systems have explored automated precision farming (3), supply chain logistics (demand forecasting and waste reduction) (4), food processing and manufacturing (5), autonomous vehicles for surveillance and diagnostics of crops (6), accelerated crop breeding (7), and consumer health and nutrition (8), among other use cases. In this relatively early stage of AI integration, it is key for developers and users to consider the social, ethical, and legal impacts of the technology (9). As AI continues to shape food and agriculture, its adoption must be guided by technical innovation and by careful attention to the broader social, ethical, and legal dimensions that will determine its long-term impact.

The rapid expansion of AI-driven tools in recent years has outpaced guidelines and regulations, leaving some technologies without oversight or tethered to obsolete guardrails (10). Responsive, up-to-date, and proactive guidelines are needed to ensure the benefits and innovation of AI-driven technologies while minimizing risks and harms related to the technologies and their misuse. Key concerns about AI throughout the food value chain include workforce reskilling and upskilling, biases and accuracy, data ownership and security, and environmental impact. AI therefore raises a combination of ethical, legal, and social issues in food systems that must be addressed to responsibly manage its design, development, and deployment.

This white paper provides an overview of discussions and activities sponsored by the NSF-Responsible Design, Development, and Deployment of Technologies (ReDDDoT) program (11). These convenings included Communities of Practice and an in-person workshop to discuss the state of AI policies, ethics regarding AI in food systems, and resulting recommendations for key stakeholders, such as policymakers, AI developers, researchers, and end-users.

AI Policy and Regulation
Policy and regulation of AI is a key issue, as the technology's rapid expansion has led to a growing awareness of AI's potential for significant impact and even harm in affected sectors

(12). In recent years, agricultural and food system researchers have called for regulatory frameworks to safeguard end-users and strengthen trust in AI technologies. Regulations for AI and its applications are in development, often with differing policies at each level of government. The state of California has been developing bills related to AI for the last few years — with 30 bills being proposed in just the first quarter of 2025 (13), while at the federal level, there is currently no comprehensive legislation (14).

The adoption and acceptance of AI technology can be hindered by ineffective, uncoordinated regulatory efforts. Interviews with agriculture and food system researchers on responsible or trustworthy AI research suggest that regulatory pathways are not grounded in the realities of implementation and are usually delayed relative to the speed of advancing AI capabilities (12). The misalignment of regulation across government, which leads to stifling regulation in some cases and insufficient regulation in others, creates risk for many stakeholders who are growing in their reliance on, or exposure to, AI tools.

Ensuring Responsible AI Adoption in Food Systems

Food holds a distinctive place in society — not only as a source of nutrition, but also as a foundation of cultural traditions and social relationships. Therefore, the ethical challenges of AI in food systems are different from the challenges in other sectors without cultural connotations or societal impacts (12). Many studies have raised concerns about the purpose and applicability of AI tools in agriculture, cautioning that agricultural commodities, whether crops or livestock, could be modified to adapt to existing technology, rather than the reverse (15). This scenario is plausible as many technology companies enter the agriculture and food sector without understanding the needs and nuances of the end-users and their operations (12).

Addressing this misalignment requires deliberate strategies, such as joint visioning, co-design, and iterative collaboration with end-users and other stakeholders (16,17). Such engagement is critical to ensure AI adoption is both effective and socially responsible. Farmers, in particular, remain concerned about data privacy and fair compensation for profits generated from agricultural data provided by the farmer (18,19). Other key issues include bias embedded in training data (20), inadequate data protection laws (21), and the financial and liability risks associated with adopting new technologies (12).

To build trust and ensure AI delivers real value to the sector, a coordinated and collaborative effort is needed to collect stakeholder input, identify regulatory priorities, and establish shared principles for responsible innovation. Such efforts are essential for ensuring that AI supports, rather than undermines, the broader societal goals of sustainability, equity, and resilience in food systems.

ReDDDoT Convenings: Communities of Practice & Workshop

Overview of Programming & Scope

In the first quarter of 2025, virtual and in-person gatherings were held to identify goals for AI in US food systems that address societal challenges of sustainability, climate-resilience, and equitable access to nutritious and healthy foods. These convenings demonstrated the importance of structured, multi-stakeholder dialogue in shaping responsible AI pathways to show how participatory processes can surface challenges that technical research alone may overlook. Facilitated discussions and activities at these events were framed around three deliverables that helped identify goals for AI in US food systems:

- Identify priorities and needs for responsible and ethical AI that can address societal challenges in food systems.
- Develop recommendations for education and workforce training that can support stakeholder engagement in AI development for food systems.
- Determine best practices for collaboration and partnership-building among stakeholders across the technology life cycle of AI in food systems.

Deliverable 1: Priorities, Needs & Recommendations for Responsible & Ethical AI in Food Systems

Convenings surfaced a range of insights on the opportunities and challenges of integrating AI into US food systems. While participants recognized AI's transformative potential, they also highlighted barriers that must be addressed to ensure that AI adoption is responsible, equitable, and sustainable.

Accessibility & Equity

Barriers: A prominent concern among stakeholders was the lack of access to AI technology, particularly among small-scale food businesses (e.g., growers, processors) and those in rural areas and from marginalized communities. Limited access to AI technology can be caused by a lack of connectivity, affordability, and training and by language barriers.

- Physical isolation of stakeholders, especially those in rural areas, and limited or absent internet connection can impede the operation and maintenance of AI-driven technologies, as many of these tools require high-speed data transmission capabilities to function (22,23).
- Connectivity can intersect with the challenge of affordability in some cases, as many small-scale food businesses have limited funds, time, and other resources to invest in innovative technologies (24,25). Investment in new technology can, at the same time, represent a potential loss of income if the technology fails. However, the uncertainty of food markets and decreasing profitability for many commodities is pressuring growers, for example, to seek innovations and technology solutions that maximize yield.
- Technology adoption goes hand in hand with the availability of training for end-users, and many potential users either lack the basic education, access to training programs, or other resources pivotal to training such as time or funding (26).
- The above issues can also be compounded with language barriers, as user interfaces, training information, and regulatory guidelines are often offered only in English, despite much of the work force in food systems speaking languages other than English (27).

Solutions: The above challenges can be addressed through investments in local infrastructure and community support programs, as well as targeted product designs by technology makers. Internet connectivity in rural areas, a longstanding issue, can be improved through government programs to bring broadband to disconnected communities (28). With increased connectivity, training programs, such as those offered through the USDA's National Institute of Food and Agriculture (29), can reach more stakeholders and potential technology users. However, training programs still need to consider the diversity of technology applications, different levels of education and training needs, and linguistic barriers to training. Similarly, training manuals provided by technology developers, regulatory guidelines, best practices from trade associations, and user interfaces of the technology should be offered in multiple languages — perhaps taking advantage of AI to make accurate translations at scale (30). To navigate financial barriers to technology adoption, developers can use lower-cost hardware, offer subscription-based models for technology services, or shared equipment programs, which can reduce the individual user's financial burden related to purchases and maintenance (24).

> ***Spotlight Case Study:***
> ***Adoption & Equity***
>
> A community representative highlighted the barriers that small and underserved farmers face in adopting AI tools. They pointed out issues of affordability, connectivity, and language access, noting that many tools are designed without consideration of diverse user needs. They urged developers to co-design technologies with farmers and communities to ensure AI adoption does not widen inequities in the food system.

Trust & Transparency

Barriers: Trust remains a crucial factor in AI adoption. Concerns about data privacy, ownership, and misuse were raised across all stakeholder groups.

- Data privacy concerns center on sensitive information from technology users being shared with unknown or unintended third parties. Farmers surveyed on this topic described worries about the loss of competitive advantages and increased scrutiny by regulators (26), concerns that can easily translate to other segments of US food systems.
- Similarly, technology developers stand to profit from user data, as this information is fed into AI models that benefit more than the original user. Businesses are hesitant to give up ownership of their data for the benefit and profit of AI developers, especially if businesses are incurring high costs to use the AI-driven technology (26).

> ***Spotlight Case Study:***
> ***Trust & Transparency***
>
> One participant reflected on the risks of relying on AI-driven sensor systems for crop monitoring. They noted how sensor drift and data inaccuracies can cascade through AI models, creating misleading outputs that undermine confidence among farmers. The participant emphasized the urgent need for stronger cybersecurity and safeguards to prevent data misuse. They also highlighted that trust must be built not only through technical accuracy but also through transparency in how AI systems are designed and validated.

- Confidence in the fate of data, as well as the efficacy of technology, are critical to widespread adoption of the AI tools in food systems. AI tools can appear to be a risky investment if their benefit is not communicated, they do not address an ongoing problem, or if they conflict with longstanding practices, such as those based on traditional knowledge (31). This hesitancy may be amplified among communities with lower access to education, training, and other resources, as described above.

Solutions: Transparent communication about data use, security, and technology outcomes is critical to AI tool adoption.

- Resources, such as user-friendly and easy-to-understand data licensing agreements, can support mutually beneficial and secure practices for technology users to implement AI tools (26). However, technology developers should be more intentional about the design of their AI models, creating, from the outset, pathways to protect and honor the privacy, profitability, and competitive advantage of individual users. Trust-building initiatives, such as participatory design, peer-led demonstrations, and dissemination of local success stories, can help dispel fears and gain user input in these design and development processes.
- Communicating the economic and operational value of AI tools before full-scale deployment is also a key strategy for building credibility, one that should be considered early in the design and development process. With more investment in demonstrating the benefits of AI tools, users will be more open and trusting of the technology, especially if the tool's outcomes or actions are explainable (rather than an unknown 'black box'). Explainable AI can offer insight into how outputs are generated, without compromising competitive advantages.

Data Guidelines & Infrastructure

Barriers: To accelerate the adoption of AI in food systems, significant advances are needed in how data is generated, managed, and applied. One of the most pressing challenges is the availability of high-quality, validated, longitudinal, and localized training data for AI models. Food systems are inherently diverse, shaped by variations in climate, soil, crop types, cultural practices, and market dynamics. Without regional time-series data, AI models risk producing outputs that are inaccurate, biased, or poorly suited to the realities on the ground (32,33). Building robust datasets requires coordinated efforts among end-users, researchers, government agencies, and industry partners to collect and share information in ways that are consistent, representative, and ethically sound.

Spotlight Case Study: Data & Infrastructure

One presenter spoke about the difficulties of obtaining high-quality, localized data for AI applications in food systems. They noted that datasets often fail to capture regional variability in soil, climate, and practices, leading to biased or unreliable recommendations. The participant stressed the importance of collaborative data-sharing initiatives that balance privacy with utility, so AI tools can be both scientifically rigorous and practical for end-users.

Solutions: The design, development, and deployment of AI-driven tools must align with the realities of diverse food systems contexts.

- Innovations in remote sensing, particularly when paired with low-cost technologies, can make advanced monitoring capabilities accessible to smallholder farmers who often lack the resources to invest in expensive equipment. Critically, scaling remote sensing data into accurate, validated estimates builds the trust needed for widespread AI adoption. This trust is essential not only to ensure that AI-generated recommendations are relevant and actionable in the field, but also to encourage data sharing that strengthens the collective datasets required to improve AI accuracy across the food system.
- At the same time, user-friendly AI interfaces that bridge the gap between research and on-the-ground practice are urgently needed. Simplifying navigation, using intuitive visuals, and tailoring outputs to local contexts empowers farmers and food system stakeholders to engage with AI tools confidently. This ease of use reduces technical barriers and fosters trust, while also supporting the long-term goal of generating large, diverse datasets that make AI systems more inclusive, precise, and effective

Policy & Systemic Needs

Barriers: The food and agriculture sector is a highly regulated space given concerns related to nutrition, food safety, and environmental impacts, among other risks. Developing AI-driven technology within this sector can therefore be difficult if regulations do not allow for innovative tools or approaches (10). Additionally, regulations that are created in response to new AI technologies – e.g., in a previously unregulated field – can stifle further development. Thus, regulations must be specific and rigid enough to be effective for the current status of technology, but flexible enough to allow for and apply to future developments and contexts.

Solutions: Proactive regulation is needed to support the infrastructure and navigate challenges that are linked to food systems (34). For instance, policies that regulate data privacy, data ownership, or data security can provide helpful frameworks to protect AI users. AI adoption could also be aided through policies that forge developer-user partnerships to trial innovations and expedite commercialization (35) and policies that subsidize or assist with regional equipment sharing programs can facilitate technology adoption (26).

Effective policy is the foundation of responsible and impactful AI adoption in food systems and can address many of the issues stated above. Regulatory frameworks should be proactive, providing clear, accessible guidelines that outline ethical standards, transparency requirements, and data governance principles. Such regulations must anticipate emerging challenges rather than react to them, ensuring that AI deployment is safe, fair, and accountable from the outset. Public and private funding should be directed toward priority areas, including the generation of high-quality, localized datasets for training AI models, expanding access to enabling technologies, and creating incentives for meaningful stakeholder engagement and co-design processes. Policies should also support the development of multi-language AI interfaces and training materials to ensure equitable access across languages and cultures. Finally, national and international agricultural strategies should explicitly prioritize AI topics in food production, distribution, and resilience planning, positioning the technology as a key driver of food security and sustainability.

Deliverable 2: Recommendations for Education & Workforce Training

Barriers: Large-scale adoption of AI technology requires a well-trained and receptive workforce and supportive regulations that promote innovation and responsibility. Stakeholders raised concerns that an AI-ready workforce, educational resources, and supportive regulations are generally not present across US food systems.

- As with most equipment and tools in food systems, AI-powered technology will require operation, oversight, or programming by a human user. In much of the food and agriculture sector, there is a shortage of skilled workers capable of interfacing with AI-powered technology, especially in farming, where the average education of a farmworker is at an eighth-grade level (27,36). Furthermore, the average age of farmworkers and farm owners is over 57, suggesting that even with upskilling or re-training of the current workforce, there may be large skilled labor shortages in agriculture in the near future (37).
- Educational resources are often lacking to support training and reskilling for an AI-ready workforce, especially in under-resourced regions (38). K-12 and adult education programs, for instance, may lack funding for AI-based curricula, as well as teaching staff with relevant backgrounds and expertise (39,40).

Solutions: Education is a cornerstone for enabling equitable and effective adoption of AI in food systems (41). Building AI capacity across the food value chain requires a coordinated approach to education and workforce development. At the foundational level, a general education requirement for AI in K–16 curricula can equip future generations with the skills to critically understand and responsibly use AI tools (42,43). For professionals already working in agriculture, food processing, or supply chain management, certificate and degree programs can provide in-depth training, offering the skills needed to integrate AI into operations and decision-making processes (44). At the same time, small-scale users — such as smallholder farmers or local food entrepreneurs — often face time and resource constraints that make long-term training impractical. For these groups, short and focused courses can deliver essential skills and knowledge in an accessible format, enabling them to adopt AI tools effectively without disrupting their livelihoods (45). By offering a spectrum of educational pathways, from foundational learning to advanced professional development, the food system can cultivate a workforce and user base that is both technically capable and inclusive.

Deliverable 3: Best Practices for Collaboration & Partnership Building

A collaborative approach towards design, development, and deployment of AI tools will be critical to its sustainable deployment (16). Workshop participants emphasized the importance of stakeholder engagement, highlighting the need for the following themes.

Inclusive Participation

When building AI solutions for food systems, inclusive participation is essential to ensuring that technologies are both effective and widely adopted. This means directly involving end-users — such as farmers, processors, and distributors — in the design and development process so that their priorities, concerns, and lived realities shape the design. Addressing sensitive issues, such as fears of job replacement, requires open dialogue and co-creation, where users can voice their perspectives and influence the outcomes. Participation can be strengthened through tangible incentives and flexible engagement formats that accommodate stakeholders with limited time, resources, or mobility. Hosting developer "open houses" and on-site field trials offers a practical, hands-on approach, allowing users to see AI tools in action, test their functionality, and provide immediate feedback. These opportunities not only refine the tools but also foster trust and a sense of shared ownership. Crucially, showcasing clear, evidence-based results, such as measurable improvements in yield and profit, helps stakeholders understand the value proposition of AI, making them more likely to adopt and advocate for its use within their communities.

Transparent Data Practices

Transparent data practices are fundamental to building trust in AI for food systems. End-users must be confident that their information is handled responsibly, which means ensuring data privacy and ownership through secure, anonymized collection methods that protect sensitive details while still enabling valuable insights. Trust is further strengthened by demonstrating the effectiveness of AI tools through proofs of concept, giving stakeholders the opportunity to independently verify results and see how the technology performs in real-world conditions. Transparency also extends to the development process itself: AI tools should be co-created with stakeholders, integrating their practical knowledge of agricultural systems with the technical expertise of researchers and developers. This balanced approach not only safeguards ethical data use but also ensures that the resulting tools are relevant, credible, and aligned with the needs of the communities they are intended to serve.

Tailored Communication & Support

Tailored communication and support are critical to ensuring that AI tools in food systems are usable and accessible to the widest possible range of stakeholders. User interfaces should be designed with simplicity in mind, accommodating both small-scale farmers with limited digital experience and larger-scale operators managing more complex systems. Incorporating multiple language options within the interface is essential, allowing users to engage with the technology in the language with which they are most comfortable. Beyond translation, effective communication requires sensitivity to cultural contexts and the removal of barriers that can hinder technology adoption — particularly for underserved groups. This includes using visual cues, intuitive navigation, and localized examples to make the technology more relatable and easier to integrate into existing practices. By aligning communication and support with the realities of diverse user groups, AI tools can achieve broader acceptance and more meaningful impact across food systems.

Best Practices by Stakeholder Group

For AI Researchers: Researchers should commit to transparent communication of methods, results, and limitations, ensuring stakeholders understand how AI models are developed and applied. Data-sharing initiatives should include incentives for contributions, coupled with strict protocols for anonymized and protected data collection to safeguard privacy and trust.

For AI Developers and Industry: AI tools should be co-developed with input from farmers, processors, and other stakeholders, ensuring solutions meet real-world needs. Trust-building can be fostered through field demonstrations, pilot programs, and clear evidence of how AI adoption can enhance profitability or efficiency. Flexible subscription models, including sliding-scale pricing, shared equipment, or one-time purchase options, can lower adoption barriers. Interfaces should be simplified, using intuitive design and localization features to maximize usability across diverse user groups.

For AI End-Users and Food Value Chain Stakeholders: Stakeholders should seek training opportunities to develop familiarity with AI tools and their potential applications. Direct advocacy for inclusion in AI development processes can help ensure that tools reflect on-the-ground realities. By engaging early and consistently, end-users can shape AI systems to be more relevant, equitable, and beneficial for their communities.

Translating Convenings into Practice

The ultimate value of this input on AI and food systems depends on whether recommendations and best practices are implemented and sustained. Gathering recommendations via multi-stakeholder engagement is an important first step, but it must be accompanied by mechanisms to measure and track adoption, impact, and lessons learned. Technology needs, capabilities, and impacts, along with stakeholder perspectives, will change over time. Accordingly, policymakers and regulators must establish continuity structures, such as evaluation frameworks, reporting requirements, open comment periods and town halls, that consider the evolution of AI and food systems and the current regulatory landscape. Embedding these learnings into regulatory and funding processes will create accountability, enable iterative improvements, and drive measurable, long-term change and benefits to stakeholders across food systems.

Conclusion

The findings and recommendations presented here are not exhaustive but represent high-level concerns and priorities from a broad group of stakeholders in the AI and food systems sectors. As technology like AI rapidly evolves, the integration of that technology across the food value chain should be co-development with stakeholders and continually re-evaluated to ensure positive outcomes for all. Priorities, needs, funding, and policy should evolve proactively as new and anticipated concerns, capabilities, and questions about the technology arise. Ultimately, responsible use of AI in food systems will depend on coordinated action across policymakers, funders, industry, researchers, and end-users to ensure the technology delivers fair, sustainable benefits that strengthen both US communities and the food system.

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